

VIVADO-AI-ASST (v1.0)

Course Specification

Course Description

Explore how to use the AMD Vivado™ AI Assistant Extension to streamline and accelerate FPGA and adaptive SoC design workflows. This course introduces AI-assisted methodologies for design creation, analysis, timing closure, and debugging within the Vivado IDE environment. Participants learn how to leverage MCP-based knowledge grounding and guided prompts to obtain reliable, reproducible results. The course focuses on how AI can improve productivity and enable efficient RTL-to-bitstream development and hardware validation workflows.

The emphasis of this course is on:

- Explaining the role and benefits of using the Vivado AI Assistant in FPGA design workflows
- Describing MCP architecture and knowledge grounding for trusted AI interactions
- Demonstrating AI-assisted project creation, block design, and IP integration
- Applying AI-driven design analysis techniques, including linting and log analysis
- Performing multi-run comparisons and identifying optimization opportunities
- Implementing timing closure strategies using iterative AI-guided workflows
- Debugging simulation, AXI protocol, and hardware issues using AI-assisted methodologies

Level – FPGA 1

Course Details

- 1 days ILT (approx. 5 hours)
- 2 lectures
- 5 labs

Course Part Number – VIVADO-AI-ASST

Who Should Attend? – FPGA and adaptive SoC design engineers, and anyone who wants to evaluate AI-assisted design workflows

Prerequisites

- Basic knowledge of AMD FPGAs and adaptive SoCs
- Familiarity with Vivado Design Suite
- Understanding of RTL design and simulation concepts

Software Tools

- Vivado Design Suite 2025.2
- Vivado AI Assistant Extension
- Visual Studio Code (VS Code)
- GitHub Copilot (for AI-assisted workflows)

Hardware

- Demo board: Versal™ VCK190 Evaluation Platform and Kria™ KV260 Vision AI Starter Kit

After completing this comprehensive training, you will have the necessary skills to:

- Explain how the Vivado AI Assistant integrates into FPGA design workflows
- Set up and validate an AI-enabled Vivado development environment
- Create and implement designs using AI-guided workflows
- Analyze design data using linting, logs, and multi-run techniques

- Apply structured timing closure methodologies using AI assistance
- Debug simulation and protocol issues (e.g., AXI) using guided workflows
- Perform hardware debug using AI-assisted ILA and VIO insertion

Course Outline

- **AMD Vivado AI Assistant: Introduction**
Describes the AI-assisted design workflows, MCP architecture, and trust models {Lecture}
- **Agentic AI Workflows for Advanced Adaptive SoC Design**
Provides a deep dive into MCP, agent capabilities, and knowledge-based assistance {Lecture}
- **Vivado AI Assistant Knowledge Base Exploration**
Explores the Vivado AI Assistant through IDE setup verification and knowledge base interaction, highlighting practical workflows and citation-driven insights. {Lab}
- **Spec-driven Design Creation Prototype on the Kria KV260 Vision AI Starter Kit**
Uses VS Code prompts to build, implement, and validate a complete Vivado design on the Kria KV260, including custom RTL integration and DMA-based hardware verification. {Lab}
- **Vivado AI Assistant Design Analysis**
Uses the Vivado AI Assistant via VS Code to perform RTL linting, debug optimization issues, and compare implementation strategies in a structured timing-closure workflow. {Lab}
- **AXI4 Protocol Debug and AI-driven Timing Closure**
Explores a two-stage design closure workflow using Vivado AI Assistant, covering AXI protocol debug in simulation followed by iterative timing closure and validation. {Lab}
- **AXIS-ILA and AXIS-VIO Insertion Flow on the Versal VCK190 Evaluation Kit**
Explores building and instrumenting a PL-only AXI4-stream debug pipeline on Versal VCK190 using Vivado AI Assistant, focusing on AXIS-based runtime monitoring and JTAG-driven observation. {Lab}

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